

Sex and Gender in Kidney Health and Disease

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KEY POINTS

- There is sexual dimorphism in the kidney, with differences in kidney physiology and disease linked to sex-specific autosomal gene expression, sex chromosome complement, and the biological actions of sex hormones.
- Evidence on the risk and outcomes of acute kidney injury in women and men is mixed and dependent on etiology.
- Women have a higher prevalence of chronic kidney disease (CKD) stages 3–5, while men may have a higher prevalence of CKD stages 1–2 due to a greater prevalence of albuminuria.
- Men experience a faster decline in kidney function and are more likely to initiate kidney replacement therapy.
- Women with CKD are less likely than men to be aware of their condition, screened, diagnosed, referred to nephrologist care, and referred for kidney transplantation.
- Evidence informing kidney care of transgender and gender-diverse persons is limited, and a holistic approach considering sex assigned at birth, sex hormone configuration, gender identity, and consideration of direct glomerular filtration rate measurement at key clinical decision points is recommended.

DEFINITIONS OF SEX AND GENDER

Kidney health and disease differ by sex¹ (biological factors) and gender¹ (sociocultural factors).^{2–5} The underlying pathophysiology, presentation, and outcomes of kidney disease and its complications, as well as the choice of, adherence to, and response to therapy, differ by sex and gender (Table 22.1). Sex, although often referred to as a binary concept (e.g., female and male), encompasses a continuum and reflects sex-related biological differences in genetics, physiology, immunology, and anatomy affecting kidney health and disease. Gender is a complex concept composed of expression (e.g., how one presents oneself), identity (e.g., transgender or cisgender girl/woman, boy/man, gender-diverse, and nonbinary), roles (e.g., caregiver and primary earner), relations (e.g., how individuals interact

with one another based on their own gender identity and the perceived gender identity of another), and institutionalized gender (e.g., the distribution of power and resources throughout society). While sex and gender are terms that are commonly used interchangeably, they each represent distinct yet interacting concepts that influence kidney and other health outcomes.⁴ Furthermore, these terms are often conflated and used in a binary fashion,⁶ resulting in a paucity of literature informing care of transgender, nonbinary, and gender-diverse persons living with kidney disease.^{7,8} To avoid assumptions, the terminology used throughout this chapter reflects the terms used in the cited literature.

Sex differences in kidney disease epidemiology, presentation, complications, and outcomes can stem from biological variations differences. However, differences can also arise due to the sociocultural attributes of masculinity and femininity (e.g., gender) whereby men, women, and

Table 22.1 Sex- and Gender-Related Factors That Contribute to Differences in the Epidemiology of Kidney Disease in Men and Women

Difference	Sex-Related Factors	Gender-Related Factors	Refs.
Men have a higher prevalence of albuminuria and thus CKD stages 1 and 2	On average, women have lower creatinine generation for the same level of urine albumin as men, resulting in higher ACRs and overestimation of albuminuria Men have a higher prevalence of diabetes mellitus and hypertension than women	Men have a higher prevalence of smoking than women Men may receive more frequent testing for albuminuria in routine health care	35,37-48
Men experience faster loss of kidney function	Protective effects of estrogens and damaging effects of testosterone on kidney health Sex-specific effects of risk factors for kidney damage including BMI and hyperglycemia	Greater prevalence of unhealthy lifestyle behaviors in men, such as more frequent smoking, poorer diet quality, and difficulty in adhering to dietary restrictions In men, increased metabolic burden and requirement for clearing of nutrients by the kidney may promote hyperfiltration and accelerate kidney damage Men are more likely to be screened and monitored for CKD and referred to a nephrologist (selection bias)	44-46,63-71,76-82, 91-95,102-104
The prevalence of CKD stages 3-5 is higher in women	Lower mean body surface area of women than men, resulting in underestimation of eGFR using creatinine-based equations Longer life expectancy of women due to biological factors	Healthier lifestyle behaviors in women that contribute to longer life expectancy and more time at risk of developing CKD Less overall screening, monitoring, and fewer referrals of women in primary care might result in a greater proportion of screening of women at risk and therefore identification of a higher proportion of women with CKD stages 3-5 (selection bias)	28,29,31,32,41-46, 91,92,102-104, 178,179
Men are at a higher risk of KRT	CKD progression is faster in men	Elderly women are more likely than men to opt for conservative care Clinicians may put more emphasis on treating CKD in men	29,45,46,74,75, 102-104,118,119
Men are more likely to be waitlisted for transplantation	Higher levels of preformed antibodies in women may be a barrier to waitlisting	Obesity and older age are greater barriers to transplant waitlisting in women	141-145,155
Living kidney donors are more often female than male	Greater prevalence of morbidities in men	Donor pool health is related to gender Altruism, pressure to donate, and financial barriers	146,149-152
Complex sex and gender differences exist in allograft survival	Estrogens may drive poorer allograft survival in women, particularly those aged 15-24 years In female (but not male) recipients, immune reactions to H-Y minor histocompatibility antigens present on male allografts may explain their poor graft survival	Adherence to immunosuppressive medications is greater in women than men	155,156,158
Women are less likely to receive arteriovenous fistulas as first vascular access for hemodialysis	Potentially smaller vascular diameters in women (despite a lack of strong evidence that such a difference exists)	Cosmetic reasons and personal choice	124,129-131,183
Women undergoing dialysis require more erythropoietin medication	Women have lower hemoglobin levels than men	CKD guidelines provide the same hemoglobin target for women and men	132,133,137-139

Table 22.1 Sex- and Gender-Related Factors That Contribute to Differences in the Epidemiology of Kidney Disease in Men and Women (Continued)

Difference	Sex-Related Factors	Gender-Related Factors	Refs.
Women experience more adverse drug events than men	Differences in the pharmacokinetics and pharmacodynamics of medications	Women are more likely to be prescribed potentially inappropriate nephrotoxic medications Women are often prescribed higher doses in relation to their body weight	109-113

ACR, Albumin-to-creatinine ratio; *CKD*, chronic kidney disease; *eGFR*, estimated glomerular filtration rate; *KRT*, kidney replacement therapy. Adapted with permission from ref.³

gender-diverse individuals may be treated differently or cope with their disease in a different way, which may be further shaped by cultural and social behavioral norms and expectations. Together, biological, social, and behavioral differences across individuals can lead to variations in disease prevalence, progression rates, and treatment outcomes (see Fig. 22.1).⁹

SEXUAL DIMORPHISM IN THE KIDNEY

In general, male kidneys tend to be larger, with hypertrophy in the proximal tubules, higher mitochondrial content, and different expressions of transporters.¹⁰ Studies in rats and mice have shown that females excrete the same amount of sodium at a lower arterial pressure than males.^{11,12} Sex differences also exist in the relative abundance of transporters along the tubule,¹³ which may impact susceptibility to nephrotoxic exposures. Sodium transporter profiles along the renal tubules vary between males and females in both mice and rats.¹⁴ In females, the expression of sodium/hydrogen exchanger 3, the primary transporter in the proximal renal tubule, is lower compared with males. Conversely, in the distal segments of the renal tubule, the abundance of sodium/chloride cotransporter and epithelial sodium channel is higher in female rats than in male rats.¹⁴

Female kidneys generally have a smaller relative weight compared with male kidneys, accompanied by a lower total nephron count.¹⁵ In female children, glomeruli are notably larger than those in age-matched male children.¹⁶ The influence of sex hormones during kidney development may play a role in shaping future kidney health.¹⁷⁻²⁰ For instance, in female mice treated with testosterone, the kidneys exhibited increased weight, primarily due to cortical thickening resulting from hypertrophy of the glomeruli and convoluted tubules.²⁰ In nephrectomized rats, male remnant kidneys showed a significantly higher growth rate than those of females.¹⁷ While this could be influenced by testosterone, tracking renal mesangial cell proliferation revealed no statistically significant effect from testosterone, while estrogen had a modest effect on proliferation and suppressed total collagen synthesis.¹⁸

SEX CHROMOSOMES

The sex chromosome complement can influence gene transcription independently of sex hormones, resulting in

age-specific, sex-specific, and tissue-specific variations in gene expression. Differences in hypertension and kidney disease risk between females and males can partially be attributed to differential expression of sex chromosome genes.²¹ A number of genes responsible for inherited kidney disorders are located on the X chromosome.²² Since males possess only one active X chromosome, these disorders tend to manifest more frequently in male individuals. However, the clinical presentation of X-linked kidney conditions in female carriers is increasingly recognized.²² Both the sympathetic nervous system and the renin-angiotensin-aldosterone system (RAAS) are regulated by sex chromosomes and hormones. For instance, the sympathetic neurotransmitter tyrosine hydroxylase is upregulated by the *Sry* gene, which has been linked to hypertension in male spontaneously hypertensive rats.²³ Genes encoding angiotensin type 2 receptor and ACE2 are located on the X chromosome, so an increased dose of these genes may contribute to the relative cardiorenal protective effects seen in females.²⁴ Both estrogen and the X chromosome push the RAAS toward the cardiorenal protective depressor pathways, which counter the classical actions of angiotensin II at the predominant angiotensin type 1 receptor, in females.²⁵⁻²⁷ In contrast, the Y chromosome and testosterone shift the RAAS balance toward the prohypertensive pressor arm, which may contribute to increased risk of kidney disease in males.²⁷⁻²⁹

SEX-SPECIFIC AUTOSOMAL GENE EXPRESSION

Sex differences in global gene expression in the F344 rat kidney across the lifespan have been identified, with specific periods of sexually dimorphic gene expression linked to functional categories such as xenobiotic metabolism, immune cell activity, and inflammatory responses.³⁰ Multiomics approaches combining RNA-seq, ATAC-seq, and proteomics to examine gene expression, chromatin accessibility, and protein expression in the proximal tubules of male and female mice have revealed numerous sex-biased genes and proteins linked to kidney functions, including metabolic and transport processes.¹⁰ In wild-type and ERbetaKO mice, the kidney had the third largest number of genes regulated by E2.³¹ Hundreds of genes are expressed differently between males and females, and these differences can be tissue specific.^{31,32} For example, genes for renal sodium transporters show differential expression between male and female rats in an age-dependent manner,³³ which

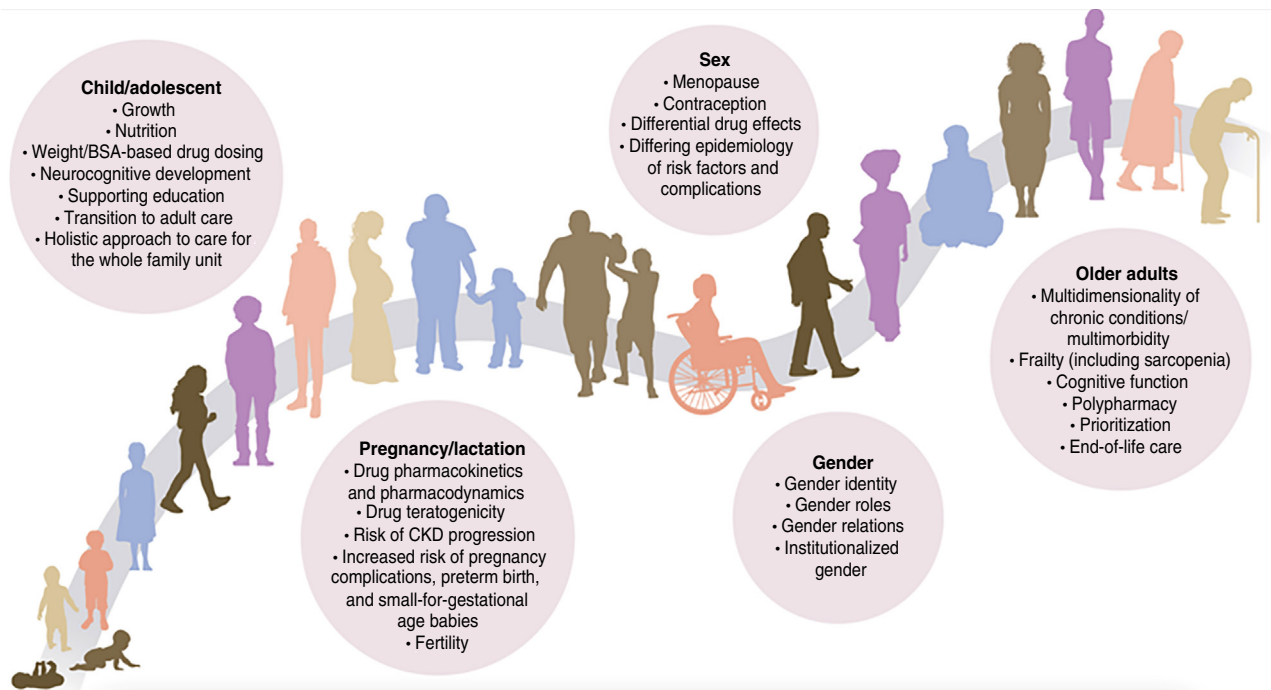


Fig. 22.1 Special considerations for chronic kidney disease care across the lifespan. BSA, Body surface area. (Reprinted with permission.⁹)

may contribute to sex differences in pressure natriuresis and long-term blood pressure regulation and risk of kidney disease.^{14,34} For example, the proximal tubule in female rats showed greater phosphorylation of Na⁺/H⁺ exchanger isoform 3 (NHE3), distribution of NHE3 at the microvilli base, and blunted expression of Na⁺/Pi cotransporter 2, claudin-2, and aquaporin 1 compared with the proximal tubule of male rats. The distal nephrons of female rats have been shown to have a greater total and phosphorylated Na⁺/Cl⁻ cotransporter (NCC), claudin-7, and cleaved forms of epithelial Na⁺ channel (ENaC) α and γ subunits.¹⁴

Overall, the lower fractional Na⁺ reabsorption observed in females is likely due primarily to their smaller transport area and lower NHE3 and claudin-2 expression levels. Notably, unlike most Na⁺ transporters whose expression levels are lower in females, Na⁺-glucose cotransporter 2 (SGLT2) expression levels are 2.5-fold higher in females.³⁵ Model simulations indicate that the increased SGLT2 expression in females may help compensate for their smaller tubular transport area, allowing them to achieve a level of hyperglycemic tolerance comparable with that of males.³⁵

SEX HORMONES AND KIDNEY HEALTH

Sex differences exist in kidney physiology and disease, underpinned by the biological actions of sex hormones.⁵ The kidney is a key regulator of sex hormones in persons living with CKD.^{36,37} Hormonal effects drive differences in kidney physiology between males and females. Differences in vascular, inflammatory, antioxidant, and other cytoprotective pathways that modify risk of kidney disease and its consequences may relate to baseline sex differences or adaptations to different exposures or injury. However, the exact influence of sex hormones on kidney function and disease remains elusive.

TESTOSTERONE

Animal studies have consistently demonstrated that testosterone can impair kidney function.^{38,39} For example, in vitro experiments showed that testosterone-induced renal tubular epithelial cell death in a dose-dependent manner.³⁹ Similarly, castration in males increases renal clearance, reduces proteinuria, and protects against glomerular injury, as observed through renal morphology assessments.^{40,41} However, experimental data in humans remain limited. In humans, genetically predicted serum testosterone concentrations have been associated with reduced kidney function in men.⁴² A clinical case report indicated that testosterone therapy led to reduced renal function.⁴³ A randomized controlled trial of 48 men reported that 6 months of testosterone treatment resulted in a decrease in eGFR.⁴⁴ Conversely, some observational studies have suggested that higher testosterone levels are associated with lower mortality in patients with chronic kidney disease (CKD)⁴⁵ and a higher eGFR.⁴⁶ A systematic review and meta-analysis examining the associations between sex hormones and the risk of cardiovascular disease and mortality in male and female patients with CKD reported that lower total testosterone concentration was associated with a higher risk of cardiovascular mortality (hazard ratio [HR] 4.37 [95% confidence interval {CI} 1.40–13.65]) and all-cause mortality (1.96 [1.35–2.83]) in male individuals with CKD.⁴⁷ The association between testosterone and these same outcomes in female individuals with CKD was not reported.

ESTROGEN

Estrogen appears to stimulate, while testosterone impairs, superoxide dismutase (SOD) responses, resulting in sex differences in oxidative stress activity with acute kidney injury.^{48,49} Basal superoxide production is significantly higher in female SOD3^{-/-} mice compared with their

wild-type counterparts, but this difference is not observed in males. The loss of SOD3 impairs reperfusion following renal ischemia in male mice, but not in females, likely due to the preservation of total SOD activity in females.⁴⁸ In male rats, orchiectomy accelerated the postischemic activation of manganese SOD and reduced reactive oxygen species and lipid peroxidation, resulting in reduced kidney susceptibility to ischemia/reperfusion injury.⁴⁹

Estrogen stimulates the release of nitric oxide (NO) resulting in vasodilation.⁵⁰ NO deficiency is associated with acceleration of kidney injury through reduction of vasodilation and endothelial dysfunction. Experimental studies demonstrate that administration of continuous estradiol can prevent glomerulosclerosis and albuminuria.⁵¹ Estrogen promotes the stability of the glomerular endothelial barrier in mice in the setting of ischemia/reperfusion AKI.⁵² However, these findings from *in vitro* and animal studies have not been directly applicable to humans.

While endogenous estradiol may offer renoprotective benefits in women,⁵³ the effect of exogenous sex hormones on kidney function remains more controversial. Research investigating the effects of both endogenous and exogenous estrogen in the form of contraception and menopausal hormone therapy on kidney health and disease has sometimes yielded conflicting results.^{53a,54-59} This likely reflects multiple important factors, such as timing of initiation of hormone therapy relative to menopausal onset, route of administration, concomitant use and type of progestin, and duration of hormone use. It is also possible that the indication for use of postmenopausal hormone therapy in humans plays a role in kidney risk, as vasomotor symptoms have been associated with altered endothelial function.⁶⁰

Two studies have examined the association between circulating estradiol concentrations and the risk of cardiovascular and all-cause mortality in female patients with CKD. A U-shaped association between serum estradiol concentrations and the risk of cardiovascular (HR 5.13 [1.29–20.3] and 4.21 [1.17–15.1]) and all-cause mortality (HR 4.49 [1.59–12.6] and 4.32 [1.59–11.7]), for the lowest and highest tertiles of estradiol, respectively, was reported in postmenopausal female individuals treated with hemodialysis.⁶¹ Another study reported that higher estradiol level was associated with greater risk of all-cause, but not cardiovascular, mortality in a population of premenopausal and postmenopausal-aged female individuals with kidney failure treated with hemodialysis (HR 1.86 [1.14–3.01]).⁶² A systematic review examining the use of postmenopausal hormone therapy and cardiovascular outcomes, as well as established measures of cardiovascular risk in women with CKD, reported limited literature on hormone therapy being associated with increased HDL and decreased LDL cholesterol.⁶³

IMPLICATIONS FOR DIAGNOSIS OF ACUTE KIDNEY INJURY

The evidence for sex and gender differences in risk and outcomes of AKI in inpatient settings is conflicting and is likely related to etiology and varying definitions of AKI.⁶⁴ AKI is a heterogeneous syndrome characterized by a rapid loss in GFR over hours or days leading to ischemia, tubule epithelial cell death and derangement, endothelial cell loss

and damage, immune cell infiltration, and inflammatory responses to the cellular debris and hypoxia.⁶⁵ Diagnosis and classification of AKI severity variably use the Kidney Disease: Improving Global Outcomes (KDIGO), Risk, Injury, Failure, Loss, and End-stage Kidney (RIFLE), and Acute Kidney Injury Network (AKIN) criteria.⁶⁶⁻⁶⁸ The use of differing AKI definitions, coupled with the fact that no definition takes sex into account, results in challenges estimating the sex-specific incidence of AKI. Furthermore, baseline levels of creatinine are generally lower in women than in men, which has implications for the sex-specific incidence of AKI as diagnostic criteria for AKI are often based on a relative and/or absolute increase in serum creatinine from a baseline value. Because of sex differences in the rate of creatinine generation and elimination as well as in volume of distribution, different criteria to define AKI could result in different sex-stratified incidence rates. When AKI is defined by a percentage change in the level of serum creatinine, the absolute change in creatinine needed to qualify as an AKI event is lower in women than in men. Conversely, when AKI is defined by an absolute increase in serum creatinine level, the percentage change in serum creatinine required to qualify as an AKI event is greater in women than in men.⁶⁹ Furthermore, the sensitivity of diagnosis codes to identify a 100% change in serum creatinine level during hospital admission has been shown to be significantly greater in men than in women.^{70,71} Of note, studies applying the KDIGO criteria appear to identify more men compared with women with AKI than those applying the RIFLE criteria.^{72,73} The 33rd Acute Disease Quality Initiative consensus statements on the role of sex and gender in AKI summarized critical gaps and opportunities to enhance the rigor of both preclinical and clinical research by better addressing both sex and gender.⁶⁴ AKI is discussed further in [Chapters 27 and 28](#).

IMPLICATIONS FOR CHRONIC KIDNEY DISEASE PREVALENCE AND ESTIMATION OF KIDNEY FUNCTION

Population-based studies demonstrate that the epidemiology of CKD stages G3–G5 varies widely across countries by sex, and overall CKD affects more women than men, especially with regard to stage G3 CKD ([Fig. 22.2](#)).³ Sex differences in CKD prevalence may represent actual differences in CKD prevalence. The simplest explanation is that the higher female prevalence of CKD is due to women's greater longevity relative to that of men combined with the natural age-related GFR decline.^{74,75} However, there is geographic variation in the difference in CKD prevalence between women and men, and these differences may also be attributable, at least in part, to heterogeneity in data collection periods, variability in equations used to estimate GFR, issues with creatinine assay calibration, non-GFR determinants (e.g., differences in meat intake or muscle mass), and biases in population selection.⁷⁶

Of note, it is also possible that the use of estimated GFR equations overdiagnoses CKD in women because equations such as the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) or the Modification of Diet in Renal Disease (MDRD) study equations were not developed in community-based populations with wide age and racial/

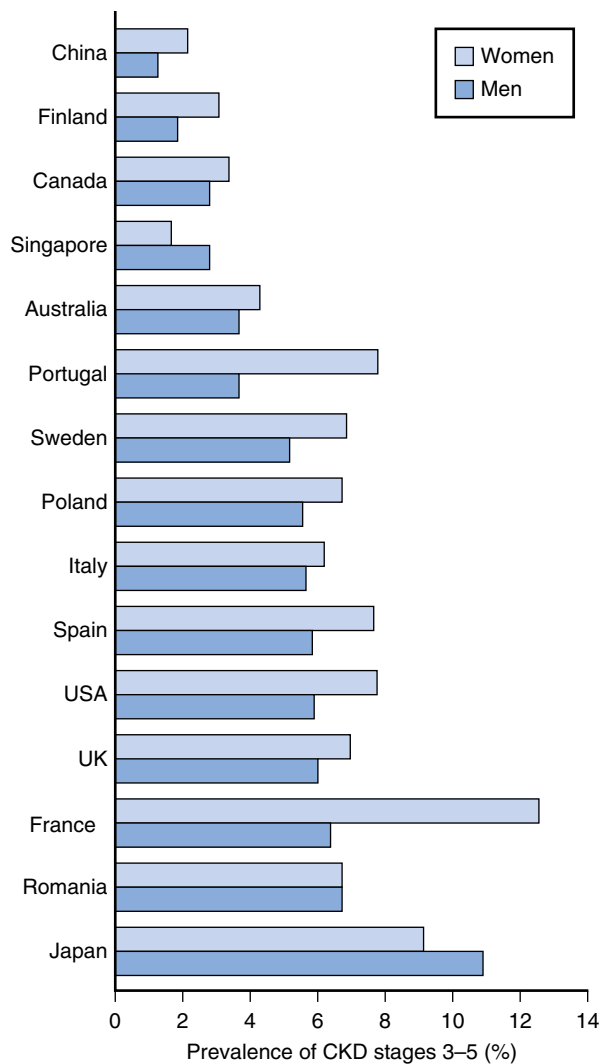


Fig. 22.2 Prevalence of chronic kidney disease stages 3–5 in men and women. (Reprinted with permission from ref³.)

ethnic diversity. In a study of 294 unselected healthy and racially diverse participants (48% women), measured GFR (mGFR) using plasma clearance of iohexol uncorrected for body surface area (BSA) was lower in women compared with men (mean difference 21.39 mL/min; 95% CI 26.75–16.03 mL/min); however, this difference was largely attenuated after correction for BSA (mean difference 9.34 mL/min; 95% CI 13.53–5.15 mL/min).⁷⁷ Furthermore, comparing the performance of mGFR employing plasma clearance of iohexol with that of eGFR using the 2009 CKD-EPI creatinine-, cystatin C-, or a combination of creatinine and cystatin C-based equations demonstrated that eGFR underestimated mGFR in women more often than in men (mean difference of mGFR-eGFR_{crea} 14.2 mL/min; 95% CI 16.5–10.9 mL/min in women vs. 3.4 mL/min; 95% CI 6.3–0.0 mL/min in men).⁷⁸ These findings suggest there could be a consistent underestimation of eGFR in women in population-based studies, resulting in a higher CKD prevalence observed in women compared with men.

CKD stages 1 and 2 account for the largest portion of the CKD population, and while data on albuminuria to determine sex-based differences in CKD stages 1–5 is limited,

albuminuria may be more prevalent and more severe in men. European population-based studies have reported higher proportions of men with CKD stages 1–5 compared to women (7.5% vs. 6.5% and 12% vs. 9.5%, respectively), largely driven by a higher incidence of albuminuria in men, although results from studies on the prevalence of CKD stages 1 and 2 in men and women are conflicting (Fig. 22.3).^{78a}

There are limitations in studies examining the prevalence of albuminuria, however. The higher prevalence of albuminuria in men may be partly explained by higher age-standardized rates of diabetes, hypertension, and smoking compared with women. Importantly, many of these studies rely on health care database data, which may not differentiate between differences due to biologically sex-based factors and those due to gender-related considerations, such as disparities in health care access, gender biases in disease recognition, or a lack of sex considerations when setting risk thresholds. For example, some studies indicate that men are more likely to undergo albuminuria testing than women in routine health care.^{79,80} Additionally, women generally produce less creatinine and have lower urinary creatinine concentrations than men,⁸¹ leading to higher urinary albumin-to-creatinine ratios at the same level of urine albumin.⁸² As a result, the current practice of using the same albuminuria categories (e.g., sex-agnostic) for defining KDIGO stages of CKD in all individuals may overestimate the prevalence of albuminuria in women. This would suggest that actual sex differences in prevalence of CKD stages 1–5 may be even larger than indicated by current data.

KIDNEY HEALTH AND CARE OF TRANSGENDER AND GENDER-DIVERSE INDIVIDUALS

Equations used to estimate eGFR commonly use a sex/gender covariate, although in the case of an individual in whom sex assigned at birth and gender identity differs, it is unclear which variable results in a closer estimation of true GFR. Substantial differences in eGFR equation performance exist between transgender men and transgender women treated with gender-affirming hormone therapy (GAHT), although the performance of CysC-based compared with Cr-based eGFR equations may be less dependent on whether sex assigned at birth or gender identity is used in the calculation.⁸³ A systematic review and meta-analysis to characterize the change in serum creatinine, other kidney function biomarkers, and GFR in transgender persons initiating masculinizing and feminizing GAHT reported that after 12 months of GAHT initiation, there was an increase in serum creatinine in transgender men but no associated change in serum creatinine in transgender women.⁸⁴ In a cohort of transgender individuals treated with GAHT for 12 months, cystatin C-based eGFR increased with estradiol and antiandrogen therapy and decreased with testosterone therapy.⁸⁵ A single-center cross-sectional study of inpatients reported a lower prevalence of AKI and CKD in transgender individuals receiving GAHT compared with those not receiving GAHT.⁸⁶ In a prospective cohort of 23 individuals treated for three months with feminizing GAHT, mGFR and kidney perfusion increased (+3.6% and +9.1%, respectively; $P < 0.05$) without increased glomerular pressure. In 21 individuals treated with masculinizing GAHT, mGFR and kidney perfusion remained unchanged after three months of treatment.^{86a} At present, a holistic approach, taking into

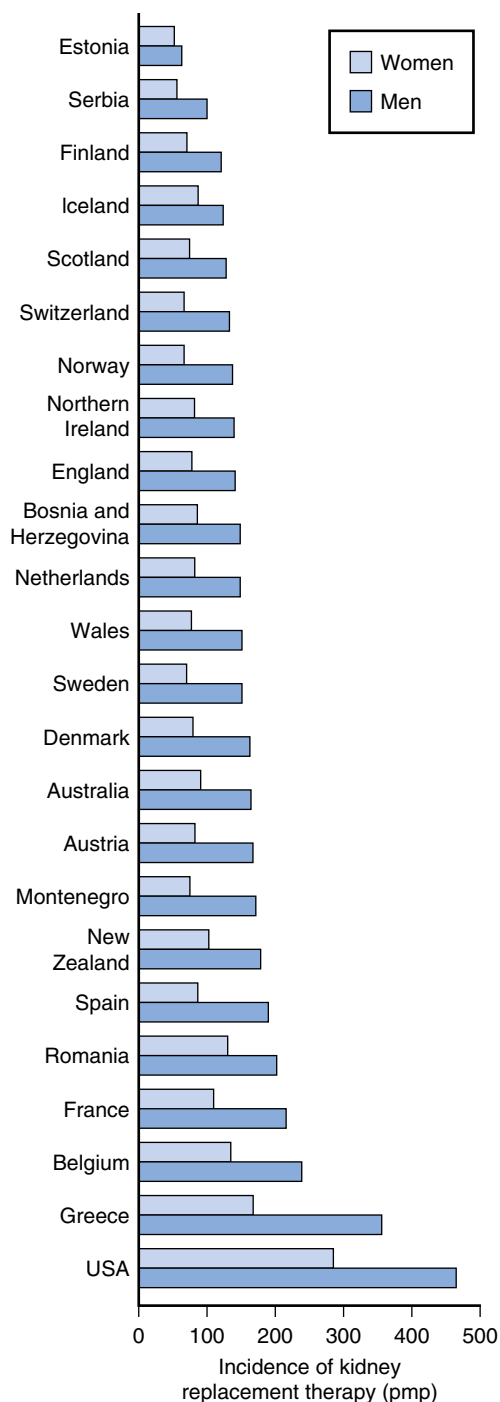


Fig. 22.3 Prevalence of CKD stages 1 to 2 in men and women. (Reprinted with permission from ref.³)

account the person's sex assigned at birth, sex hormone configuration (e.g., GAHT use and/or gonadectomy), and gender identity, with shared decision making between the individual living with or at risk of kidney disease and the provider, and measuring GFR directly at important clinical decision-making points is advised.^{87,88}

GLOMERULAR FILTRATION RATE DECLINE

Men have higher measured GFR compared with women but appear to have faster decline with age.⁸⁹⁻⁹⁶ Data from

population-based studies suggest that kidney function declines faster in males (i.e., steeper slope of eGFR decline) than in females.⁹⁷ For example, the Dutch PREVEND study reported a mean eGFR slope of -0.55 ± 1.47 mL/min/1.73 m² per year in men compared with -0.33 ± 1.41 mL/min/1.73 m² per year in women.⁹⁸ Multiple sex-related factors likely contribute to this faster GFR decline in men. Males have increased activation and tissue responsiveness of the renal RAAS.⁹⁹⁻¹⁰¹ The male renal vasculature becomes increasingly nitric oxide-dependent with age compared with the female renal vasculature¹⁰²; thus any kidney disease interfering with nitric oxide production could cause CKD to progress more quickly in males compared with females.

Among healthy Norwegian adults, the age-related decline in iohexol-measured GFR (mGFR) was faster in men than women.³ Between the ages of 50 and 72 years, women had a lower mGFR than men, but by a mean age of 72 years, women had a higher mGFR due to a slower rate of kidney function decline. This observation aligns with the higher prevalence of CKD stage 3 in women and the greater incidence of dialysis and kidney transplantation in men.³ Of note, individuals with major chronic diseases (including diabetes, cardiovascular disease, or kidney disease) or CKD risk factors (e.g., smoking) were excluded from this study, suggesting the difference in mGFR decline between men and women may be more reflective of biological or sex-related mechanisms rather than lifestyle behaviors, although this remains speculative.

PROGRESSION OF CHRONIC KIDNEY DISEASE

Previous studies have examined differences between women and men in CKD progression, with conflicting results. First, a meta-analysis examining the risk of CKD progression in 68 cohort studies of patients with nondiabetic CKD concluded that men progress to kidney failure faster than women.¹⁰³ Subsequently, a patient-level meta-analysis of 11 randomized trials using angiotensin-converting enzyme (ACE) inhibitors in patients with CKD concluded that the rate of CKD progression was more rapid in women compared with men.⁵³ Another patient-level meta-analysis of 46 cohort studies (2,051,158 participants [54% women]) from Europe, North and South America, Asia, and Australasia from general population ($n = 1,861,052$), high-risk ($n = 151,494$), and CKD ($n = 38,612$) cohorts showed no evidence of a sex difference in associations of eGFR and urinary albumin-creatinine ratio (ACR) with kidney failure risk.¹⁰⁴ These conflicting conclusions are likely due to several factors including the mixed nature of included studies (e.g., population-based studies vs. referral to nephrology, observational vs. randomized studies), differences in populations (e.g., primarily premenopausal¹⁰³ vs. postmenopausal⁵³ women), and where some studies report loss of eGFR and others report kidney replacement therapy (KRT) initiation as the primary outcome. Kidney failure as an endpoint can be problematic due to competing risks, as death is more likely to occur than progression to kidney failure, and men die earlier than women. Moreover, older women tend to choose conservative management rather than KRT compared with men.^{105,106}

Higher albuminuria in men compared with women may explain their faster progression to kidney failure. A general population cohort reported that men were 3.45 (95% CI 2.23–5.33) times more likely than women to experience

worsening of albuminuria.¹⁰⁷ In another general population cohort, higher albuminuria at baseline was associated with a decline in eGFR in men but not in women with 7-year follow-up data.¹⁰⁸ Adjusting for albuminuria nullified any sex difference in the rate of eGFR loss or progression to dialysis in patients referred to nephrology care in Sweden, suggesting that albuminuria may be a mediator of the higher risk in men.⁸⁹ Men had a 50% higher risk of kidney failure than women in a pooled analysis of four Italian observational cohort studies of patients with moderate to advanced CKD, which may be partly attributed to higher levels of proteinuria in men.¹⁰⁹ In a cohort of patients with diabetic kidney disease referred to nephrology, men had greater albuminuria and a more rapid decline in eGFR compared with women; albuminuria predicted rapid CKD progression in men but not women.¹¹⁰

Gender factors may also play a role in the greater risk of CKD progression observed in men compared with women with CKD. While men are more frequently prescribed ACE inhibitors and statins,^{79,111} they also exhibit higher rates of behavioral cardiovascular risk factors that influence risk of CKD, such as smoking and alcohol consumption, and are more likely to have poorer dietary habits. In contrast, women tend to adopt primary cardiovascular prevention strategies more readily.¹¹²

CAUSES OF CHRONIC KIDNEY DISEASE

The prevalence of the causes of CKD differs by sex. The lower rate of congenital abnormalities of the kidney and urinary tract among females may help to explain the lower incidence of CKD among females compared with males in the adolescent population.¹¹³ Diabetes mellitus is the leading cause of CKD in both men and women, affecting a similar proportion of each group in the non-dialysis-dependent stage.^{114,115} Hypertension, a well-established risk factor for cardiovascular disease, also contributes to the development and progression of CKD to kidney failure in both men and women.¹¹⁶ Blood pressure is higher in men than women, but women have an accelerated age-related rise in blood pressure that begins around the fifth decade of life¹¹⁷ with increasing cardiovascular disease risk beginning at lower thresholds of SBP for women than for men.¹¹⁸ Whether this also applies to CKD risk is unknown. CKD is discussed further in [Chapter 50](#).

Other specific causes of CKD appear to be more prevalent in either men or women. While lupus nephritis is more frequent in women, all other primary glomerular diseases are more common in men, although the sex ratios may vary at different life stages.¹¹⁹⁻¹²¹ While ANCA-associated vasculitis (AAV) glomerulonephritis may have a slight male predominance in younger populations, older patients (aged >75 years) are more likely to be female, particularly those with myeloperoxidase (MPO)-ANCA vasculitis.¹²² Approximately 20% of AAV risk appears to be genetic, with sex-specific genetic risks identified, particularly in microscopic polyangiitis (MPA).^{123,124} Lupus and ANCA vasculitis are discussed further in [Chapters 31 and 32](#).

KIDNEY STONES

The relationship between sex and the risk of kidney stones is complex.^{125,126} Men are more likely to be affected compared

with women with previous studies demonstrating a male-to-female ratio of 1.5:1 to 3:1 around the world,^{127,128} although this gap may be decreasing.¹²⁹ Some factors may pose a different level of risk for men compared with women, as indicated by varying relative risk magnitudes for waist circumference¹³⁰ and phytate,¹³¹ vitamin C,¹³² and vitamin D intake.¹³³ The disparity between women and men may be partially explained by lifestyle risk factors; a significant portion of the excess risk can be attributed to differences in urine chemistries.¹²⁵ While a U.S. national cross-sectional study reported no independent association between sex hormones (testosterone and estradiol) and history of kidney stones in either males or females, women with low estrogen may still be at clinically significant increased risk of developing stones (OR 1.47, 95% CI 1.00, 2.16).¹³⁴ Menopause has been associated with kidney stones, although no association was found between menopausal hormone therapy use and risk.¹³⁵

PREECLAMPSIA

A systematic review and meta-analysis of observational studies found that women with a history of adverse pregnancy outcomes had significantly higher risk of long-term kidney disease, with the risk of kidney failure being highest among women who experienced preeclampsia.¹³⁶ A systematic review and meta-analysis examining the risk of CKD after preeclampsia reported a significant increase in risk of kidney failure after preeclampsia (meta-analytic risk ratios [95% confidence interval] 6.35 [2.73–14.79]), while the risk of albuminuria and CKD increased but statistical significance was not reached (4.31 [0.95–19.58] and 2.03 [0.58–7.32], respectively).¹³⁷ Pregnancy and kidney disease are discussed further in [Chapter 58](#).

DIAGNOSIS AND MANAGEMENT OF CHRONIC KIDNEY DISEASE

An analysis of data from the 1999 to 2000 U.S. National Health and Nutrition Examination Survey (NHANES) reported that women with CKD were less likely than men to be aware of their condition.¹³⁸ Among those with eGFR in the 30 to 59 mL/min/1.73 m² range, only 2.9% of women, compared with 17.9% of men, were aware they had CKD. A subsequent analysis almost 2 decades later of the 2018 NHANES data found that, despite an overall improvement in CKD awareness, the gender gap persists, with women at a disadvantage.^{138a} In this study, “CKD awareness” was self-reported based on the question, “Have you ever been told by a health care professional that you had weak or failing kidneys?” This suggests that differences in health care access or the receipt of care may contribute to the sex/gender-based disparity in CKD awareness, although the reasons for this disparity in awareness between women and men are unclear.

A 2022 Swedish study examined whether the care processes for patients with at least one eGFR measurement below 60 mL/min/1.73 m² aligned with guideline recommendations over a 10-year period.⁷⁹ Women had lower rates of CKD recognition (diagnosis), disease monitoring (e.g., remeasurement of creatinine or albuminuria), nephrology referrals, and the use of antiproteinuric medications in primary care compared with men irrespective of level of eGFR or albuminuria, age, and

underlying conditions. Among those meeting the criteria for nephrology referral, women were 42% less likely to visit a nephrologist within the next 18 months. A retrospective analysis of adult patients at a care network affiliated with a U.S. academic medical center reported significant sex differences in primary care-based CKD management, with females overall receiving worse care than males.⁸⁰ In the United Kingdom, men are referred to nephrologists at higher eGFR levels than women,¹³⁹ and women are less likely to be diagnosed with CKD.¹⁴⁰ In Canada, women attending primary care practices had less frequent measurements of urinary albumin-to-creatinine ratio than men.¹⁴¹ In the United States, women with incident albuminuria were less likely than men to initiate renin-angiotensin system inhibitor therapy.¹⁴²

In France,¹⁴³ Austria,¹⁴⁴ and in the CKDopps prospective study cohort (France, Brazil, Germany, United States),¹⁴⁵ more men than women receive outpatient treatment at CKD clinics, despite women having a higher prevalence of CKD stages 3–5 in these regions. A qualitative interview study of nephrologists suggested that clinicians may be more focused on men and their symptoms, while women may prioritize the health of their families over their own well-being.¹⁴⁶

COMPLICATIONS OF CHRONIC KIDNEY DISEASE

Although there is increasing recognition of sex- and gender-related divergence in disease course and complications,¹⁴⁷ to-date guidelines for the management of CKD and its complications remain sex and gender blind. The current evidence landscape for sex-based differences in risk factors and complications of CKD is outlined in Fig. 22.4.

ANEMIA

A 2020 systematic review and meta-analysis reported that female CKD patients were 36% more likely to develop anemia compared with male patients.¹⁴⁸ Observational studies report that females on ESA therapy have lower hemoglobin levels, require more erythropoietin-stimulating agents (ESA), and have a higher Erythropoietin Resistance Index (ERI) compared with males independent of iron status, comorbidities, and dialysis adequacy.^{149,150} In an international mixed-methods study of female individuals with CKD, the majority reported heavy menstrual bleeding ($n = 12$, 86% of dialysis; $n = 46$, 94% of nondialysis CKD; $n = 14$, 100% of transplanted), with 38% reporting receiving a blood transfusion.¹⁵¹ A systematic review and meta-analysis reported that menstrual abnormalities are common in females with CKD, although kidney transplantation may improve menstrual health.¹⁵²

CHRONIC KIDNEY DISEASE MINERAL AND BONE DISORDER

Women in the general population have higher serum phosphorus and FGF-23 than men,^{153,154} although whether this translates to sex differences in CKD-related clinical outcomes is unclear. A greater burden of bone disease in the aging general female compared with the male population is well described, and as such the development of CKD in women likely occurs where preexisting sex and gender-specific issues of bone health are already present. In a cohort study of 679,114 adults of 40 years and older (mean

age 62 years) stratified by CKD stage, the 3-year cumulative incidence of fracture (proximal humerus, forearm, pelvis, and hip) increased in a graded manner in adults with lower levels of eGFR for both men and women, although the risk in women was significantly higher compared with men.¹⁵⁵ Female sex has also been noted to be a risk factor for fracture in the setting of kidney transplantation.^{156,157}

IMPLICATIONS FOR KIDNEY FAILURE AND KIDNEY REPLACEMENT THERAPY

The incidence of KRT including both dialysis and kidney transplantation is higher among men compared with women in the United States (467 vs. 322 per million population)¹⁵⁸ and in Europe (180 vs. 93 per million population).^{159,162} This difference has been recognized in Europe since the 1960s¹⁶⁰ and applies to the most common causes of kidney failure, such as diabetes mellitus and hypertension, although is less pronounced in cases of glomerulonephritis.^{161,162} These differences are even more pronounced in lower resource settings for many social and structural reasons.¹⁶³ In the United States and Europe, >80% of both men and women begin KRT with hemodialysis.^{161,162}

Differences in the decision to initiate KRT, or type of KRT chosen, between women and men may have less of a biological basis and be more reflective of gender-related factors.¹⁶⁴ In a clinical practice audit, elderly women were almost three times as likely to opt for conservative management compared with dialysis compared with age-matched men.¹⁰⁶ In a UK single-center survey of 242 predialysis patients, choice of KRT (hemodialysis [HD], peritoneal dialysis [PD], and conservative management [CM]) was strongly influenced by personal and demographic parameters.¹⁶⁵ Demographic factors influencing KRT choice included being married (PD 95.7%, HD 53.8%, CM 41.7%; $P < 0.001$), being employed or a student (PD 33.3%, HD 11.5%, CM 0%; $P = 0.015$), and having another person living at home (PD 100%, HD 69.5%, CM 50%; $P = 0.003$). Family/home/work circumstances ($P = 0.003$) influenced the choice of PD compared with HD. Patients who had had a social services assessment in the prior 12 months or who had received private care services or disability allowance were more likely to choose CM.¹⁶⁵ Women initiate KRT at a lower eGFR compared with men,^{165a} although the reasons are unclear. It is possible that for the same level of eGFR, men may be more likely to develop complications that trigger dialysis initiation, reflecting biological (i.e., sex-based) differences. However, this may also reflect personal preference of the individual living with CKD or nephrology referral timing, in addition to health care provider practice patterns regarding dialysis initiation.¹⁶⁶

DIALYSIS MANAGEMENT

The practice of extrapolating findings based on men to women may account for differences in dialysis management and complications of CKD.^{147,167} For example, estimating dialysis adequacy using Kt/V (where K=dialyzer clearance of urea, t=dialysis time, and V=the distribution volume of urea) assumes a constant urea distribution volume across all individuals,¹⁶⁸ although female muscle mass, a surrogate for lean body mass, is typically lower than that of males. This

		Favors females	Favors males	No sex difference	Unclear
CKD epidemiology and risk factors	Diagnosis of eGFR		✓		
	Age				⌄
	Diabetes	✓		✓	⌄
	Hypertension	✓			⌄
	Obesity		✓		
	Cerebrovascular disease				⌄
	Smoking				⌄
	Socioeconomic status				⌄
Complications of CKD	CKD progression				⌄
	Cardiovascular disease		✓		
	Mineral and bone disease		✓		⌄
	Anemia		✓		⌄
	Quality of life		✓		⌄
	Cancer				⌄
	Mortality	✓			⌄
	Paediatric CKD				⌄

 Strong evidence
 Some evidence
 More evidence needed

Fig. 22.4 Current evidence landscape for sex-based differences in risk factors and complications of chronic kidney disease. (Reprinted with permission from ref.¹⁴⁷)

assumption may lead to underestimation of dialysis needs in women, potentially resulting in underdialysis.¹⁶⁹

Men are more likely than women to receive arteriovenous fistulas, which are considered the preferred vascular access approach¹⁷⁰⁻¹⁷³; women are more likely to receive dialysis through an arteriovenous graft or central venous catheter.¹⁷⁴ This disparity may be attributed to nephrologists' concerns about smaller vascular diameters in women that could complicate cannulation; however, there is no strong evidence supporting the idea that women have smaller veins.¹⁷⁵ Females tend to have slower maturation rates and experience lower rates of primary, primary-assisted, and secondary patency; require more procedures per patient to achieve maturation; and maintain fistula patency.¹⁷⁴

KIDNEY TRANSPLANTATION

Women are less likely than men to be deceased or live donor kidney transplant recipients^{176,177}; this difference may be partly explained by the fact there are more men than women who progress to kidney failure.¹⁶⁷ Women may also be more risk averse to aggressive treatment options.¹⁷⁸ Women tend to have higher panel reactive antibodies due to the immune reaction associated with pregnancy,¹⁷⁹ and the prevalence of human leukocyte antigen immunization increases with parity.¹⁸⁰ However, gender disparities in access to kidney transplantation are perceived by nephrologists to be exacerbated by gender

norms and values, stigma and prejudice, and educational and financial disadvantages.¹⁸¹ While women are more likely to be living kidney donors,¹⁷⁶ men are more likely to be deceased kidney donors.¹⁸² Compared with men, women with kidney failure are 45% less likely to discuss kidney transplant with their health care provider; this gap only widens with advancing age.¹⁸³ In participants aged 66 to 75 years, women had 29% less access to kidney transplant than men, and women older than age 75 years had 59% less access to transplant than men of the same age. The distribution of KRT treatment modality is illustrated in Fig. 22.5. Transplantation is discussed further in Chapter 69.

WAITLISTING

Several studies have shown that women with kidney failure have 10% to 20% less access to a kidney transplant than men, even after adjustment for demographic and clinical characteristics.¹⁸⁴⁻¹⁹² This extends to the pediatric population, where girls are less likely to receive a preemptive kidney transplant than boys.¹⁹³ Physicians and other health care providers often view women with kidney failure as being frailer than men,¹⁹⁴ which may account for lower referral rates for transplantation. Across multiple studies from the United States, Austria, Australia, and Germany, older men were significantly more likely to be placed on the transplant waiting list than older women.^{187,191,195} Furthermore, obesity disproportionately affects women's chances of being waitlisted for a transplant.¹⁹⁶ Data from the Australian and

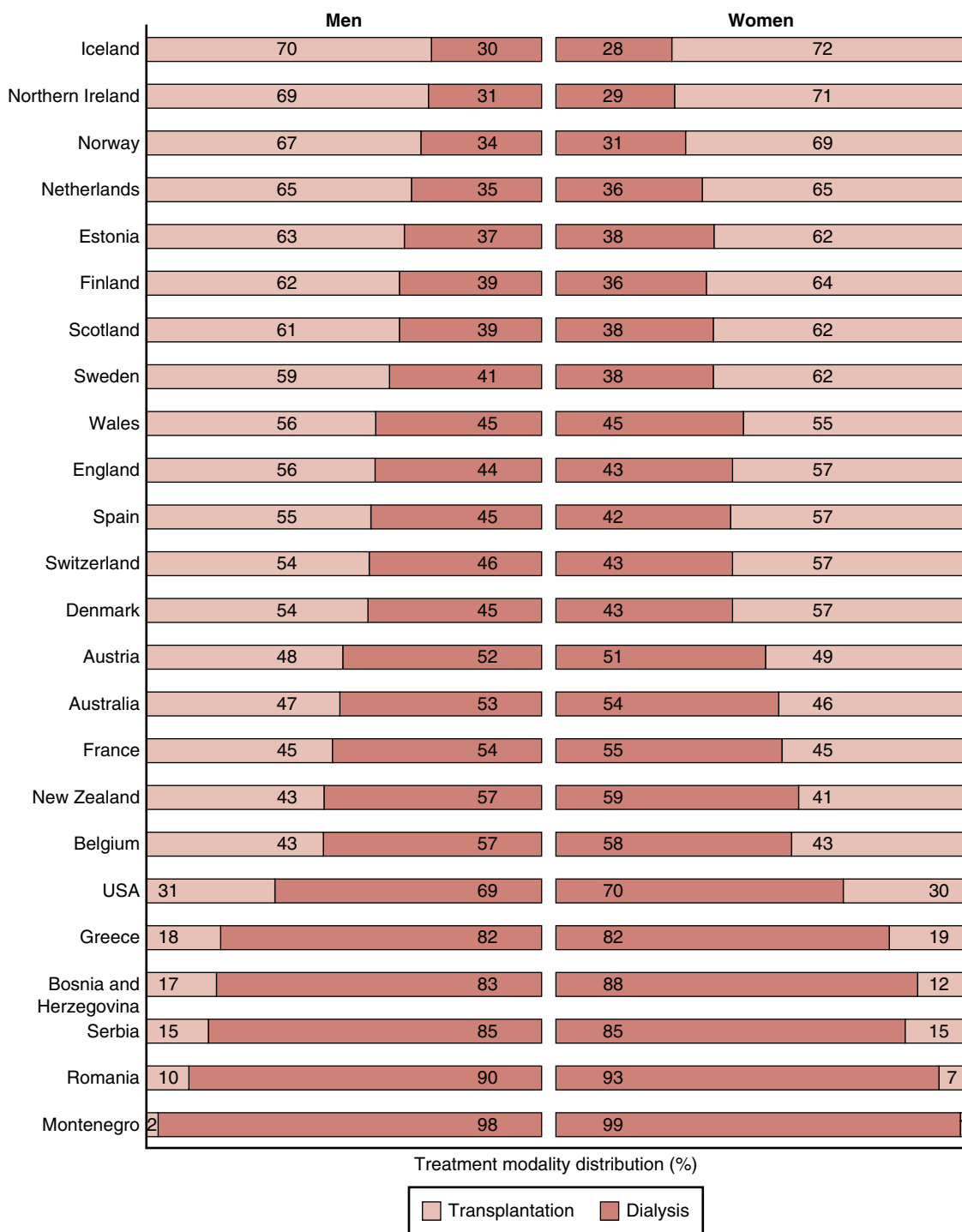


Fig. 22.5 Distribution of kidney replacement treatment modality in men and women. The portions of men and women on kidney replacement therapy who have kidney transplants or are receiving dialysis in various European countries,²⁴⁸ the United States,²⁴⁹ Australia, and New Zealand.²⁵⁰ The proportions of patients who have kidney transplants are similar in men and women. (Reprinted with permission from ref ³; Boenink R, Astley ME, Huijben JA, et al. The ERA Registry annual report 2019: summary and age comparisons. *Clin Kidney J.* 2021;15:452–472; U.S. Renal Data System. 2022 USRDS Annual Data Report: epidemiology of kidney disease in the United States. <<https://adr.usrds.org/2021>>;2022; and Australia and New Zealand Dialysis and Transplant Registry. ANZDATA Registry. 44th Report, Chapter 1: Incidence of Kidney Failure with Replacement Therapy. <<http://www.anzdata.org.au>>;2021.)

New Zealand Dialysis and Transplant Registry (2007–2014) revealed that women with obesity were 34% less likely to be waitlisted than women with ideal weight. In contrast, obese men were only 14% less likely to be waitlisted compared with men of ideal weight.¹⁹⁷

KIDNEY DONATION

According to U.S. Renal Data System data, women are more likely than men to indicate willingness to donate their organs when deceased,¹⁹⁸ with deceased women donors likely to be older and die as a result of cerebrovascular disease.^{199,200}

However, in the majority of countries kidneys are more frequently retrieved from male than female deceased donors,¹⁸² which may be attributed to a higher number of men than women being admitted to intensive care units due to trauma or stroke, which can lead to irreversible acute brain injury.²⁰¹ Women represent 6 of every 10 living kidney donors.^{176,202} Although there is no conclusive evidence why women donate more than they receive,²⁰³ this phenomenon is likely related to sociocultural factors,^{204,205} which in part are related to women's more traditional role as the caregiver in the family, with women donating first as mothers, followed by as spouses.²⁰⁶ Additionally, men are often seen as the primary earners in society, and if they become living donors, it can result in economic loss and create a financial burden for the entire family.^{203,207} Living donation is discussed further in [Chapter 70](#).

MORTALITY AND ALLOGRAFT SURVIVAL

Women have improved survival after transplantation compared with men,^{161,187} although there is conflicting and inconclusive data when comparing graft survival outcomes between women and men. Some studies have reported worse outcomes for women compared with men,^{208,209} while others have found similar outcomes,²¹⁰ and some have identified male sex as an independent factor for poorer outcomes in kidney transplant recipients.²¹¹ Of note, the effect of sex on allograft survival is affected by age.²¹²

Women have a more robust cellular and humoral immune response than men,¹⁷⁹ which is partially influenced by the effects of sex hormones on immune activation. This enhanced immune response may contribute to lower allograft survival in women. The stronger immune reaction may lead to more acute rejection episodes, resulting in allograft injury and eventual loss. Interestingly, while higher sensitization rates in women are often attributed to previous pregnancies, studies in children and young adults have also shown poorer outcomes in preadolescent girls,^{213,214} although the reasons for this remain unclear. Of note, studies that report a survival advantage for women's allografts often cite better adherence to follow-up care and immunosuppressive medication regimens in women compared with men.^{215,216}

There is increasing evidence that the sex of both the donor and the recipient are relevant to the outcome of the kidney allograft. Female allografts have been reported to have higher rates of acute rejection and allograft loss.^{216,217} These observations may be explained by factors such as a lower number of nephrons in the female kidney²¹⁸ and higher expression of human leukocyte antigens, which can cause increased immunogenicity.²¹⁶ However, in a large international study using data from the Collaborative Transplant Study, female recipients had higher excess mortality than male recipients at all ages except 45 to 59 years, and sex differences in excess mortality were statistically significant only when the donor was male.²¹⁹ A systematic review sought to evaluate the prognostic effects of recipient sex and, separately, gender as independent predictors of graft loss, death, cancer, and allograft rejection following kidney or simultaneous pancreas–kidney (SPK) transplantation and reported very low to low certainty evidence to suggest there are

no differences between male and female kidney and SPK transplant recipients.²²⁰

PATIENT-REPORTED OUTCOMES

Women consistently report poorer health-related quality of life (HRQOL) and more symptoms than men, both in the general population^{221,222} and among those with chronic disease.²²³ Similar trends are observed in patients with non-dialysis-dependent CKD and those on dialysis.¹⁶⁷ Whether these sex differences are due to CKD or are larger than in the general population remains unclear. While both men and women report improved HRQOL after transplantation, the improvement is smaller in women, indicating a CKD-specific sex difference.^{224,225} Additionally, graft loss negatively impacts HRQOL more in women than men.²²⁶ In older (≥ 65) patients with advanced CKD, men had higher HRQOL scores than women at baseline, but HRQOL declined more rapidly in men over time.²²⁷ This difference was partly due to faster eGFR decline in men. In the year before dialysis, HRQOL declined more quickly in men, but this decline stabilized more in women after the first year of dialysis.²²⁸ These findings suggest that while women report worse HRQOL overall, men with kidney disease experience a greater HRQOL decline over time. Women with CKD report more severe symptoms than men.^{229,230} Additionally, women on hemodialysis experience higher rates of anxiety and depression,¹⁷⁰ which negatively affects both mental and physical HRQOL²³¹ and may explain some of the gender differences in HRQOL among hemodialysis patients.²³² In an exploratory study,²³³ conventional hemodialysis was associated with improved Kidney Disease Quality of Life 36 physical component score in female participants, while incremental hemodialysis was associated with improved mental component score in female participants with more roles and responsibilities traditionally ascribed to women, highlighting that both sex and gender-based factors likely play a role in HRQOL in persons with CKD.

SURVIVAL

The female survival advantage persists across the full range of eGFR, though the gap narrows at lower levels.^{234,235} Once on KRT, survival is similarly poor for both women and men.^{161,162} In an Australian and New Zealand population-based cohort study using registry data linkage,²³⁶ few differences in outcomes were found between male and female patients with kidney failure treated with either dialysis or kidney transplantation. However, compared with the general population, female patients had greater excess all-cause mortality than male patients (female patients: standardized mortality ratio 11.3, 95% (CI), 11.2–11.5, expected deaths 1781, observed deaths 20,099; male patients: 6.9, 6.8, to 6.9 95% CI, 6.8–6.9, expected deaths 4272, observed deaths 29,277), particularly among younger patients and those who died from cardiovascular disease. Relative survival was also consistently lower in female patients, with adjusted excess mortality 11% higher (95% CI, 8%–13%). Average years of life lost was 3.6 years (95% CI, 3.6–3.7) greater in female patients with kidney

failure compared with male patients across all ages, with no differences by sex between those treated with hemodialysis or PD. Kidney transplantation reduced but did not entirely eliminate the dialysis-associated sex differences in excess mortality, with similar relative survival and the years of life lost difference reduced to 2.3 years (95% CI, 2.2–2.3) between female and male patients.²³⁶

Little is known about sex or gender differences in experiences of kidney failure treatment or long-term survival in pediatric populations. In a population-based cohort study of all children aged 18 years or younger with treated kidney failure in Australia (1980–2019) and New Zealand (1988–2019), 394 (20%) individuals died over 30,082 person-years of follow-up (median follow-up, 13.1 years).¹⁹³ Most children (74%) were transplanted after initiating dialysis; however, male patients were almost twice as likely as female patients to receive a preemptive living donor kidney transplant.¹⁹³ Mortality rates were 12.2/1000 person years (pys) (95% CI, 10.6–14.0/1000 pys) in male patients and 14.3/1000 pys (95% CI, 12.4–16.5/1000 pys) in female patients, with variation by age, and female patients were found to lose ≥ 7 more life years than male patients irrespective of age at diagnosis.¹⁹³

CAUSE-SPECIFIC MORTALITY

Sex differences in cause-specific mortality vary by age, treatment modality, dialysis vintage, diabetes status, and cause of death. In American adults receiving maintenance hemodialysis over a 20-year period, survival rates were similar between women and men, with median survival times of 2.72 (IQR, 1.16–5.21) and 2.69 (IQR, 1.17–5.19) years, respectively.²³⁷ Overall, women had a 9% and 15% higher likelihood of infection and withdrawal-related mortality compared with men, respectively. Conversely, women had a 7% and 10% lower likelihood of cardiovascular- and cancer-related mortality, respectively, compared with men. Of note, women (vs. men) aged 18 to 44 years had higher likelihood of excess mortality across all specific causes (including a 40% increased risk of withdrawal compared with men of the same age), while women (vs. men) aged >75 years had a lower likelihood.²³⁸

However, younger women and women with diabetes on dialysis had lower life expectancy and higher mortality than men, particularly from noncardiovascular causes,^{162,239,240} mainly due to infection-related deaths.^{239,241} Women on PD may also face a higher risk of sepsis and peritonitis-related deaths than men.²⁴²

An ERA Registry study linked higher noncardiovascular mortality in women on dialysis or with kidney transplants to a greater prevalence of multisystem disease, particularly systemic lupus erythematosus, and more frequent use of catheters for vascular access in women.^{145,172,173,239} An ANZDATA study confirmed that women had an 8% higher risk of all-cause mortality in the first 5 years after dialysis initiation, mainly due to higher rates of infection-related deaths and dialysis withdrawals.²⁴¹ Women are also more likely to die from dialysis withdrawal, possibly due to more elderly women living alone without caregivers.^{241,243,244}

In patients older than 45 years of age initiating dialysis, men, especially those without diabetes, face a higher risk of cardiovascular mortality, mainly from myocardial infarction and bleeding.²⁴⁰ After age 40, men are more likely to die

from malignancies than women, similar to trends in the general population.²³⁹ This may partly explain why women with kidney transplants have higher life expectancies than men, although the survival advantage is still smaller than in the general population.^{93,161,162}

CARDIOVASCULAR EVENTS AND MORTALITY

A systematic review and meta-analysis showed men were at marginally higher risk of cardiovascular mortality than women among the CKD population, with borderline significance.²³⁴ A pooled analysis of more than 2 million participants showed that men had higher cardiovascular and all-cause mortality at all eGFR levels, but the risk of cardiovascular mortality increased more steeply in women as eGFR declined.¹⁰⁴ In early CKD stages, women are at lower cardiovascular risk than men,²⁴⁵ but this difference diminishes at lower eGFR levels. For instance, in a Swedish study of 30,000 CKD stages 3–5 patients, cardiovascular mortality was 20% higher in men than women, but no sex difference was found in stage 5 nondialysis patients.⁸⁹ These findings suggest that the cardioprotective effect of female sex decreases as CKD progresses.^{234,235,246,247}

CONCLUSION

In summary, experimental models have identified sex differences in kidney outcomes, even when the same level of insult and functional injury is achieved. However, clinical studies often conflate sex with gender and overlook the potential confounding effects of changes in sex hormone status, behaviors, exposures, and access to care throughout the lifespan. Opportunities exist to enhance the rigor of both preclinical and clinical research by better addressing both sex and gender factors, ultimately improving kidney care for all.

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